

# Implementation and Evaluation of a Three-Stage Text-Based Person Identification System

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## Abstract

Person identification—figuring out who someone is from camera data—is vital for surveillance, autonomous vehicles, and security. Most existing methods rely on facial recognition or pre-collected images, which are vulnerable to spoofing and fail when no prior photos are available (for example, in criminal investigations). In addition, deploying such systems in real-world environments requires real-time, parallel processing to handle multiple video streams efficiently. To address these challenges, we present a practical and scalable person identification system applied on dash camera input in real-world scenarios. First, the system performs object detection and multi-object tracking on each video stream to extract non-facial features—such as clothing and accessories—and crops those relevant frames. Second, the cropped frames are fed in parallel into a cross-modal module that converts text descriptions into image-feature embeddings, enabling effective text-based person retrieval. Our system demonstrates strong real-time performance and retrieval accuracy, highlighting its potential for deployment in real-world, dash camera-based person search applications. It fills a critical gap in non-facial, cross-modal retrieval and industrial-scale, real-time multi-stream processing for dashcam scenarios, in sharp contrast to existing Text-Based Person Retrieval and Re-ID methods that focus on fixed-camera setups. Future work may explore incremental modeling of clothing changes, RGB-T fusion, and privacy-compliant deployment.

## CCS Concepts

• **Computing methodologies** → *Visual content-based indexing and retrieval.*

## Keywords

Object tracking, Person Re-Identification, cross-modal, Object Detection, Image Processing

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## 1 Introduction

Person identification plays a critical role in public safety, autonomous systems, and law enforcement. While facial recognition has been a dominant technique in this field—supported by advancements such as EdgeFace [7], GhostFaceNets [2], AdaFace [9], PartialFC [3], and MagFace [13]—it still suffers from notable limitations. These systems typically require clear, frontal facial images, which are difficult to obtain in real-world scenarios involving dash cameras or traffic cameras. Issues such as occlusion (e.g., by masks, helmets, or sunglasses), poor lighting, and oblique viewing angles further degrade recognition accuracy. Moreover, facial recognition systems are increasingly susceptible to spoofing techniques such as deepfakes, raising significant privacy and security concerns.

To address these challenges, we propose a person identification system that eliminates the need for facial images during the inference stage. Unlike traditional person Re-identification (Re-ID) approaches that rely on pre-collected or gallery images, our system is designed to operate even when no prior visual data of the target individual is available—enabling use cases such as suspect identification in criminal investigations or locating missing persons based on verbal or textual descriptions.

Our approach shifts the focus from face-based identification to non-facial features such as clothing, accessories, and body structure. The system integrates state-of-the-art components including YOLOv8 [8] for object detection, BoT-SORT [1] for multi-object tracking, and Transformer-based architectures [6, 14] for extracting robust visual and semantic features. By leveraging a cross-modal retrieval mechanism, we enable the matching of text descriptions to person images in a scalable and real-time manner.

To validate the effectiveness of our system, we conduct extensive experiments using real-world video streams. Key evaluation metrics such as accuracy, F1-score, and principal component analysis (PCA) are used to assess its performance. Our findings demonstrate that the proposed text-based person identification model achieves high accuracy and responsiveness, making it suitable for deployment in dynamic environments like dashcams and large-scale multi-stream processing systems.

## 2 Related Works

### 2.1 Self-Supervised Pre-Training for Transformer-Based Person Re-Identification

This work [12] investigates self-supervised strategies to pre-train transformer backbones specifically for person Re-ID. By designing pretext tasks that mimic Re-ID challenges (e.g., predicting shuffled patches, contrasting views), the model learns more robust and discriminative features before fine-tuning on labeled Re-ID datasets. Experiments show that such self-supervised pre-training significantly boosts final accuracy, especially when labeled data is scarce.

### 2.2 Pedestrian Alignment Network for Large-scale Person Re-identification

Early person re-identification (re-ID) methods relied on hand-crafted features and metric learning but struggled with misalignment from detection errors and pose variations. While deep CNNs improved accuracy, they remained sensitive to spatial misalignments. The Pedestrian Alignment Network (PAN) [16] addresses this by integrating Spatial Transformer Networks (STNs) into an end-to-end framework, adaptively rectifying feature misalignments during training. This unified approach eliminates preprocessing and achieves state-of-the-art performance on large-scale re-ID benchmarks.

### 2.3 CMTR: Cross-modality Transformer for Visible-infrared Person Re-identification

CMTR [10] is the first application of a pure transformer architecture to visible-infrared person re-identification, replacing CNN backbones with a sequence-to-sequence encoder-decoder design. It injects learnable modality embeddings into token representations to explicitly model differences between visible and infrared inputs. A modality-aware enhancement loss uses these embeddings to tighten feature clusters of the same identity and push apart different identities across modalities. The framework includes a modality compensation module that recovers missing modality information by translating features between visible and infrared domains. Extensive evaluations on SYSU-MM01 and RegDB demonstrate that CMTR outperforms state-of-the-art CNN-based methods, showcasing transformers' effectiveness in bridging the cross-modality gap.

## 3 System Design

This study presents a real-world deployment of a scalable person identification system utilizing dash camera inputs. The system is organized into three distinct stages—detection tracking, feature extraction, and cross-modal retrieval—which together deliver several key advantages. First, by isolating object detection and multi-object tracking into Stage 1, we ensure robust and consistent ID assignment across varying viewpoints and lighting conditions. Stage 2's parallelized extraction of non-facial features (clothing, accessories, etc.) via a ViT-based Image Encoder maximizes throughput and isolates appearance cues from background noise. Finally, Stage 3's cross-modal module aligns these rich image embeddings with text descriptions in real time, enabling intuitive, natural-language person searches even when facial data is missing or obscured. This

three-stage design not only boosts modularity—allowing each component to be optimized or replaced independently—but also guarantees real-time performance across multiple streams and flexible deployment in dynamic, unstructured environments, distinguishing it from approaches tied to static galleries or fixed-camera setups.

### 3.1 Three-Stage Video Processing and Modeling

This study designs a system capable of processing images from various surveillance cameras, considering a wide range of practical applications. The system architecture is divided into three main stages, as illustrated in Figure 1.

First, raw video footage is processed using YOLOv8 to detect individuals. Subsequently, Bot-SORT is employed to track these individuals and assign unique IDs. Frames tagged with IDs are then extracted and compiled into separate video frames for each ID. These segmented video frames are then fed into the parallelized Image Encoder of CLIP, where image features are extracted using a Vision Transformer (ViT).

Simultaneously, a Text Encoder, primarily based on Transformer architecture, is used to extract features from text descriptions corresponding to images. Finally, the study attempts to identify the most probable image-text pairs by calculating the outer product of the transposed matrices of text and image features. All these processes take place within a parallelized real-time monitoring model, where all results are annotated and stored with specific time frame and detailed metadata.

### 3.2 Training the model

This study is based on the ViT-L/14 model, and its training methodology and loss functions refer to [15].

$$\begin{aligned}\hat{\mathbf{f}}_{\text{image}} &= \frac{\mathbf{f}_{\text{image}}}{\|\mathbf{f}_{\text{image}}\|_2}, \quad \hat{\mathbf{f}}_{\text{text}} = \frac{\mathbf{f}_{\text{text}}}{\|\mathbf{f}_{\text{text}}\|_2} \\ \mathbf{L} &= \frac{\hat{\mathbf{f}}_{\text{image}} \cdot \hat{\mathbf{f}}_{\text{text}}^\top}{\tau} \\ \mathbf{Y} &= \mathbf{I}(1 - \epsilon) + \frac{\epsilon}{N - 1}\end{aligned}$$

$$\log(\mathbf{P}_{\text{image}}) = \log(\text{softmax}(\mathbf{L})),$$

$$\log(\mathbf{P}_{\text{text}}) = \log(\text{softmax}(\mathbf{L}^\top))$$

$$\begin{aligned}\mathcal{L}_{\text{image}} &= -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N Y_{ij} \log P_{\text{image},ij} \\ \mathcal{L}_{\text{text}} &= -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N Y_{ij} \log P_{\text{text},ij} \\ \mathcal{L} &= \frac{\mathcal{L}_{\text{image}} + \mathcal{L}_{\text{text}}}{2}\end{aligned}$$

This new loss function incorporates an entropy penalty to mitigate overfitting on small datasets. However, its effectiveness diminishes or even degrades performance when applied to large-scale datasets such as ImageNet [4] and COCO [11].

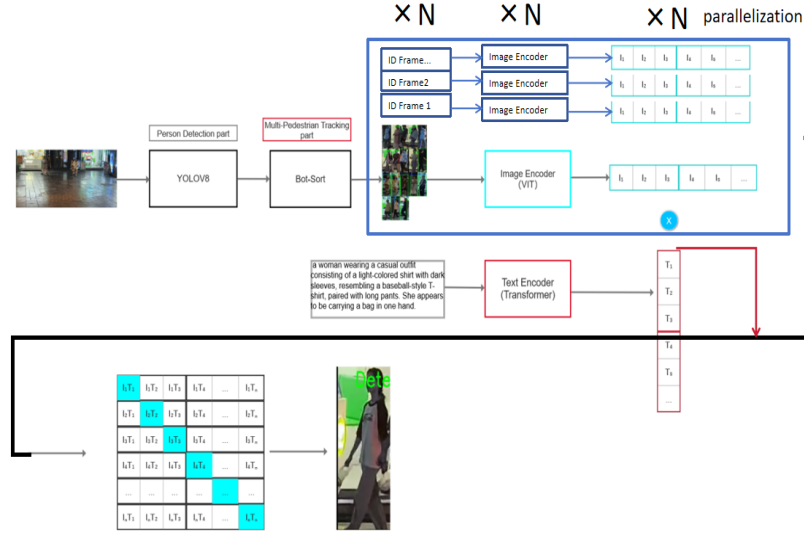


Figure 1: The overview of our System

## 4 Experimental and Evaluation

In this section, we quantitatively assess how well our proposed system meets the two core challenges outlined in the Abstract:

- (1) **Robust identification under facial-spoofing scenarios** by exploiting non-facial cues (clothing, accessories, etc.).
- (2) **Text-based person identification without pedestrian images** via cross-modality conversion (text→image).

To this end, we perform: (1) a controlled verification on the ICFG-PEDES dataset, measuring retrieval and classification performance (Section 4.1), and (2) a real-world deployment on a dash camera, evaluating end-to-end text-based identification in dynamic scenes (Section 4.3).

### 4.1 Controlled Verification on ICFG-PEDES

*Experimental Setup.*

- **Dataset:** ICFG-PEDES [5], augmented with a spoofing subset for Challenge 1.
- **Training:** Fine-tune both Image and Text encoders of CLIP with our multi-loss scheme (Figure 2).
- **Metrics:** Accuracy and F1 score for retrieval; embedding separability via PCA variance ratios.

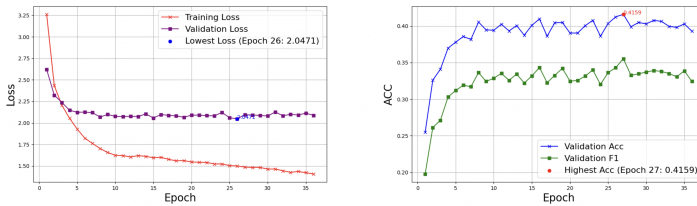


Figure 2: Loss minimization process during model training

*Results.* Figure 2 shows simultaneous convergence of image- and text-encoder losses, reaching a minimum of 2.0471 at epoch 26. Validation accuracy (Figure 3) climbs steadily, peaking at 41.59% (epoch 27), while F1 oscillates between 40% and 80%.

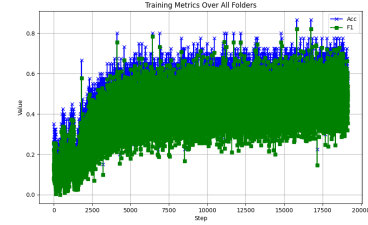


Figure 3: Validation accuracy and F1 score over training epochs

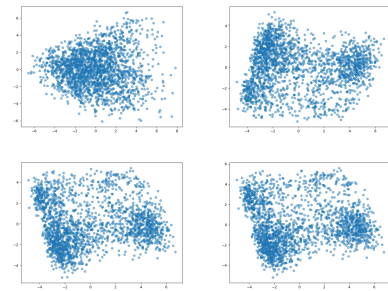


Figure 4: PCA embeddings: original CLIP vs. models at 10/20/30 epochs

Figure 4. Two-dimensional PCA projections of our joint image-text embeddings at four successive stages of training. Each scatter plot shows matching pairs (filled circles) and non-matching pairs (transparent circles) in the shared feature space. From top-left to bottom-right, the plots correspond to (a) the untrained model, (b) after the first alignment objective, (c) following coarse-grained contrastive fine-tuning, and (d) after full multi-task optimization. Notice how the matching and non-matching points gradually form tighter, more distinct clusters, demonstrating progressively improved cross-modal alignment.

## 4.2 Discussion.

- **Challenge 1 (spoofing):** Our model achieves 41% accuracy on the spoofed subset, demonstrating the viability of non-facial features for robust identification; a direct comparison with facial-only methods is left for future work.
- **Challenge 2 (cross-modality):** Text→image retrieval F1 reaches 35% by epoch 30, demonstrating effective modality bridging.

## 4.3 Real-World Dash-Camera Experiment



Figure 5: Dash-camera deployment: top-10 frames for prompt “a man with black coat and wearing a white mask”

**Setup and Results.** Figure 5. Qualitative results on real-world dashcam footage from a garbage truck. Given the text query “worker in brown jacket lifting a green bin,” our system ranks video frames by relevance and retrieves the top 10 candidates. Each thumbnail shows one of the highest-scoring frames: nine out of ten clearly depict a person wearing a brown jacket and handling a green garbage container, despite rapid camera motion, changing viewpoints, and varying occlusions. This example highlights our model’s robustness in unconstrained, dynamic environments.

## 5 Futurework

Future work will focus on enhancing the system’s practicality and scalability by addressing key challenges. First, we will next improve the model architecture to enhance its performance on cameras, optimizing it for real-time detection under real-world conditions. Second, we plan to develop a distributed system that balances processing tasks between edge devices and central servers, enabling efficient handling of high-volume video feeds with real-time synchronization. Lastly, we will optimize both server and local models, improving inference processes to reduce latency and dynamically adapt to varying network conditions, while using advanced compression techniques to minimize model size without compromising performance. These efforts will ensure the system’s effectiveness and usability in diverse applications in real world.

## 6 Conclusion

In this study, we developed a text-based person identification system by integrating advancements in computer vision, including Person Re-Identification, Cross-Modality, Multiple Object Tracking, and Object Detection. The system is designed for practical applications in the real world, aiming to bridge the gap between advanced research and real-world implementation. The task workflow was divided into three stages, enabling the processing of raw video data from cameras into a complete text-based identification system. This structured approach demonstrates the system’s capability to address complex identification tasks while highlighting its potential for practical deployment in various scenarios.

## 7 Acknowledgments

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